

Matt Flewelling

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EDUCATION

University of Edinburgh

Edinburgh, United Kingdom

MSc in Computational Applied Mathematics

2024 - 2025

- Relevant coursework: Machine Learning in Python, High Performance Data Analytics, Numerical Methods for Data, Statistical Methodology, Fundamentals of Optimisation.

Queen's University

Kingston, Ontario, Canada

BASc in Mechanical Engineering

2016 - 2021

- Dean's Scholar Award: Awarded for academic excellence, recognising top-performing students throughout all academic years.

WORK EXPERIENCE

Data Scientist

Dartmouth, Nova Scotia, Canada

Marimetrics Technologies

Oct 2022 - Dec 2023

- Contributed to the development and deployment of a web-based dashboard for visualising operational risk in aquaculture systems, including full-stack implementation and AWS hosting.
- Built cloud-based data storage and ETL pipelines using AWS Lambda, enabling near real-time analytics and decision support for IoT sensor data.
- Designed and maintained REST APIs and backend services to support scalable data access and user interaction.
- Performed exploratory data analysis on large, multi-source datasets, uncovering insights to support process optimisation and risk management.

Mechanical Engineer

Dartmouth, Nova Scotia, Canada

Marimetrics Technologies

Jul 2021 - Oct 2022

- Streamlined production and R&D workflows by implementing automation solutions for sensor calibration and testing, significantly improving throughput and consistency.
- Routinely performed ad hoc analyses to investigate anomalies and answer critical questions about environmental sensor data, supporting decision-making and system improvements.
- Designed, maintained, and extended a Python/MATLAB codebase for sensor QA, enabling robust anomaly detection and performance validation.

PROJECTS

Kalman Filter-Based Particle Tracking for Ultrasound Imaging (Dissertation)

2025

- Developed and evaluated a Kalman filter-based microbubble tracking algorithm for super-resolution ultrasound imaging using synthetic datasets with ground truth.
- Built an end-to-end MATLAB pipeline covering image pre-processing, localisation, trajectory linking, and post-processing to improve output quality.
- Benchmarked against Nearest-Neighbour and Hungarian Algorithm baselines, achieving higher robustness under noisy or uncertain inputs.
- Ran controlled experiments across varying noise levels, quantifying precision, recall, spatial error, and robustness.
- Generated flow and vessel density maps from reconstructed tracks, demonstrating trade-offs between completeness and accuracy relevant to clinical use.

Cognitive Decline Prediction (Coursework)

2025

- Built Lasso regression models to forecast ADAS13 cognitive scores using clinical and neuroimaging data.
- Engineered features using PCA, power transforms, and interaction terms; tuned models via grid search cross-validation.
- Interpreted model coefficients to assess feature influence and ensure clinical plausibility, contributing to explainability.

- Developed compartmental models (ODE-based) to estimate PET scan derived solute clearance rates in the brain using Python (lmfit, SciPy).
- Applied parameter optimisation and model comparison techniques across physiological scenarios to evaluate fit and complexity.

TECHNICAL SKILLS

Languages & Tools: Python, MATLAB, Pandas, NumPy, scikit-learn, TensorFlow, SQL.

Analytics & Experimentation: Regression, feature engineering, model tuning, statistical interpretation, EDA.

Machine Learning: Supervised learning, model development, hyperparameter tuning, deployment.

Data Engineering: AWS (Lambda, S3, DynamoDB, API Gateway), REST APIs, ETL pipelines.

Collaborative Software Development Tools: Version control (Git, Bitbucket), Jira, Confluence.

Visualisation: matplotlib, seaborn.

Experimental Design and Data Collection: Skilled in acquiring, structuring, and processing data from sensors and instruments; experience with scripting and terminal tools for ad hoc experiments.

PROFESSIONAL SKILLS

Technical Writing: Thorough documentation of R&D, synthesis of methodology, results and conclusions in R&D reporting, contributor to proposal development.

Communication of Technical Concepts: Ability to explain technical and analytical ideas clearly to both technical and non-technical audiences.

Cross-Disciplinary Collaboration: Experience working with cross-functional teams, including scientists, engineers, and product stakeholders.

Critical Thinking: Skilled at evaluating data quality, questioning assumptions, and applying logical reasoning to technical problems.

Time Management and Prioritisation: Effectively managed multiple project timelines and shifting priorities, drawing on experience in fast-paced start-up settings.

Problem-Solving and Initiative: Proactive approach to identifying inefficiencies and implementing improvements independently.